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Group actions and homogeneous spaces

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Abstract

This project provides a structured introduction to the theory of manifolds, including the generalisation of calculus in $\mathbb{R}^n$ and the basics of Lie groups and algebras. This is followed by an exploration of the connections between manifolds and group actions through homogeneous and symmetric spaces, with examples and non-examples for both.

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1 Introduction

This project explores the geometric structures that arise from symmetry, focusing on smooth manifolds, Lie groups, homogeneous spaces, and symmetric spaces. These concepts form a foundation for much of modern differential geometry and have applications in both mathematics and physics.

In Chapter 2 we begin with smooth manifolds, which generalise the idea of curved spaces where calculus still applies. Lie groups, which are both groups and smooth manifolds, allow us to study symmetry in a smooth setting. When a Lie group acts transitively on a manifold, the resulting structure is called a homogeneous space. This is the focus of Chapter 3, where we begin with its formal definition and properties, and illustrate key examples such as spheres, Grassmannians and the Klein bottle. We'll also be seeing algebraic topology properties of homogeneous spaces, and give examples of connected non-homogeneous spaces.

Among homogeneous spaces, symmetric spaces stand out for their high degree of symmetry and rich structure. They appear in many areas, including Riemannian geometry, representation theory, and physics. In Chapter 4, we define these spaces within the framework of Riemannian geometry, we explore essential properties such as geodesic completeness, and examine the decomposition of the corresponding Lie algebra. We conclude the chapter with a discussion of non-examples of symmetric spaces.

This project brings together these ideas to show how symmetry shapes geometry, using group actions as a unifying theme.

2 Background

2.1 Smooth manifolds and Lie groups

Smooth manifolds are topological spaces that are locally homeomorphic to $\mathbb{R}^n$ with a smooth structure. We will build this up ingredient by ingredient.

Definition 2.1 (Topological manifold). A topological manifold M of dimension n is a Hausdorff, second countable topological space M which is locally $\mathbb{R}^n$. In other words, for each point $p \in M$ there exists an open neighbourhood $U \subseteq M$ and a homeomorphism $\varphi : U \rightarrow V$ where $V \subset \mathbb{R}^n$ is open.

(U, φ) is called a *coordinate chart* or simply a *chart*. A manifold of dimension n is also called a n -manifold.

We have included some interesting examples violating Hausdorff and second countability in the appendix [A](#).

Definition 2.2 (Smooth function). For $U \subseteq \mathbb{R}^n$, $f : U \rightarrow \mathbb{R}^m$, we say f is *smooth* if it's infinitely differentiable.

We would like to build up smooth structures on topological manifolds to be able to do calculus. It turns out (this will be clear from many proofs in the upcoming sections) that the correct technical condition to put on charts is:

Definition 2.3 (Smoothly compatible). Let $(U, \varphi), (V, \psi)$ be charts on M , where $U \cap V \neq \emptyset$. We say they're *smoothly compatible* if the transition map

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$$

is smooth.

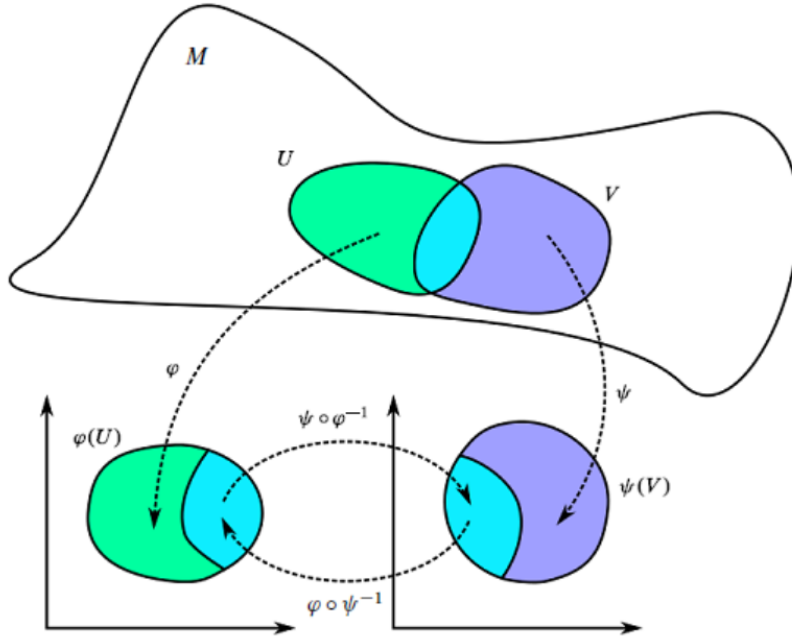


Figure 1: Diagram illustrating smoothly compatible charts and a transition map on a manifold (Hamster's Hideout, 2019).

We piece these charts (local maps) into an atlas (global map).

Definition 2.4 (Atlas). Let M be a topological n -manifold, a *smooth atlas* for M is a collection of charts $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in I}$ such that $\{U_\alpha\}_{\alpha \in I}$ covers M , and charts in $\mathcal{A}$ are pairwise smoothly compatible.

Remark 2.5. We mean a maximal atlas with respect to the partial order $\subseteq$. In fact, all smooth atlases can be extended to unique maximal smooth atlases, which can be easily proved, providing a notion of equivalent atlases.

Definition 2.6 (Smooth Structure). A *smooth structure* on a topological manifold M is a maximal smooth atlas.

Definition 2.7 (Smooth manifold). A *smooth manifold* M is a topological manifold M with a smooth structure on M .

Remark 2.8. A topological manifold may fail to obtain a unique smooth structure. Examples include $\mathbb{S}^7, \mathbb{R}^4, E_8$ (Donaldson, 1983; Freedman, 1982; Milnor, 1956).

For the rest of this paper, M is an n -manifold unless otherwise specified.

Remark 2.9. Given M and a chart (U, ϕ) on M , we usually denote $\phi(x) = (x^1, x^2, \dots, x^n)$ in a Cartesian fashion, this is called the *local representation* about p . $x^i = \pi_i \circ \phi$, where $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is the projection onto the i th coordinate, are called *coordinate functions*.

Example 2.10 (Euclidean spaces). $\mathbb{R}^n$ is a smooth n -manifold with the atlas comprising the single chart $(\mathbb{R}^n, \text{Id}_{\mathbb{R}^n})$.

Proposition 2.11. *An open subset of a manifold is itself a manifold.*

Example 2.12 (General linear group). The space of invertible matrices form an open set of $\mathbb{R}^{n \times n}$ and is automatically a manifold.

Definition 2.13 (Product manifold). Given two smooth manifolds M and N , a *product manifold* can be constructed. The topology is the same as the product topology, and for any charts $(U_\alpha, \varphi_\alpha)$ of M and (V_β, ψ_β) of N , the product map is $\Gamma_{\alpha,\beta} : U_\alpha \times V_\beta \rightarrow \varphi_\alpha(U_\alpha) \times \psi_\beta(V_\beta)$ such that:

$$\Gamma_{\alpha,\beta}(x, y) := (\varphi_\alpha(x), \psi_\beta(y))$$

We define the product atlas $\{(U_\alpha \times V_\beta, \Gamma_{\alpha,\beta})\}_{\alpha,\beta}$ and check that the transition function

$$\Gamma_{\gamma,\eta} \circ \Gamma_{\alpha,\beta}^{-1}(x, y) = (\varphi_\gamma \circ \varphi_\alpha^{-1}(x), \psi_\eta \circ \psi_\beta^{-1}(y))$$

is always smooth by $\{(U_\alpha, \varphi_\alpha)\}$ and $\{(V_\beta, \psi_\beta)\}$ being atlases. As the finite product of Hausdorff and second countable spaces remains Hausdorff and second countable, we're done.

If $\dim M = m, \dim N = n$, then the product manifold has dimension $m + n$.

Example 2.14 (Sphere). Consider the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$, and the open cover defined by

$$U_x^+ = \mathbb{S}^2 \cap \{x > 0\}, U_x^- = \mathbb{S}^2 \cap \{x < 0\}$$

$$U_y^+ = \mathbb{S}^2 \cap \{y > 0\}, U_y^- = \mathbb{S}^2 \cap \{y < 0\}$$

$$U_z^+ = \mathbb{S}^2 \cap \{z > 0\}, U_z^- = \mathbb{S}^2 \cap \{z < 0\}$$

The chart (U_x^+, φ_x) is defined by the projection $\varphi_x : U_1 \rightarrow \{(y, z) : y^2 + z^2 < 1\}$ given by $(x, y, z) \mapsto (y, z)$. The other maps are similarly projections onto unit open disks lying on planes. One can easily see how this generalise to giving a smooth structure on $\mathbb{S}^n$. See Figure 2.

Let's verify the transition maps are smooth: for instance

$$\varphi_z \circ \varphi_y^{-1}(y, z) = \varphi_z(\sqrt{1 - y^2 - z^2}, y, z) = (\sqrt{1 - y^2 - z^2}, y)$$

which is clearly smooth. The other transition maps are similarly smooth. Therefore

$$\{(U_x^\pm, \varphi_x^\pm), (U_y^\pm, \varphi_y^\pm), (U_z^\pm, \varphi_z^\pm)\}$$

is a smooth atlas, so $\mathbb{S}^2$ is a smooth 2-manifold.

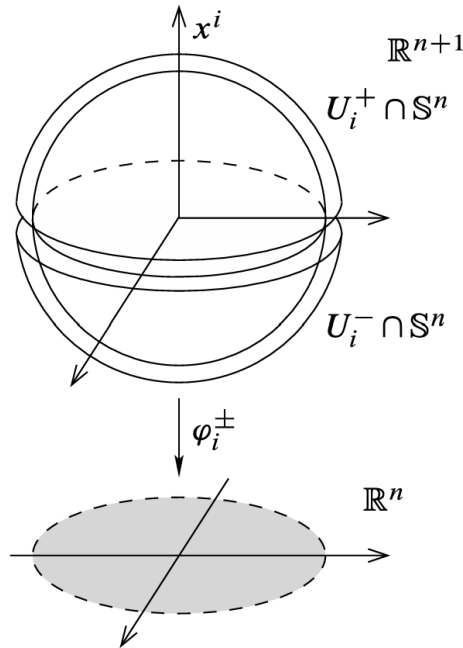


Figure 2: $n = 2$, U_3^+ and U_3^- (Lee, 2002).

Definition 2.15 (Smooth map). Let M, N be smooth manifolds. Then $f : M \rightarrow N$ is *smooth* if there exist charts (U, φ) and (V, ψ) of M and N respectively, such that $F(U) \subset V$ and

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \rightarrow \psi(V)$$

is a smooth map in the Euclidean sense.

f is a *diffeomorphism* if f is bijective and both f and $f^{-1} : N \rightarrow M$ are smooth. Then M, N are *diffeomorphic*, and we write $M \cong N$.

Remark 2.16. The category of smooth manifolds $\mathbf{Man}^\infty$ consists of smooth manifolds as objects, smooth maps as morphisms, and diffeomorphisms as isomorphisms.

Example 2.17. Consider the circle $\mathbb{S}^1$ with the standard smooth structure. The map $f : \mathbb{R} \rightarrow \mathbb{S}^1$ defined as $f(t) = \exp 2\pi it$ is smooth.

Definition 2.18 (Connected sum). Given two n -manifolds M_1, M_2 , their connected sum at points $p_1 \in M_1, p_2 \in M_2$ is constructed by the following procedure (See Figure 3):

- Take a chart (U, ϕ) of p_1 . Since $\phi(U)$ is open, we can take an open ball $\mathbb{B}_{\phi(p_1)} \subseteq \phi(U)$.
- Take a closed ball inside the open ball.
- Remove the interior of the closed ball from the manifold.
- Do the same for the other manifold. Now we have two manifolds with boundary. Identify the boundaries of the closed balls in both manifolds together.
- A chart can be defined around each point on the glued boundary in an obvious way using the original charts.

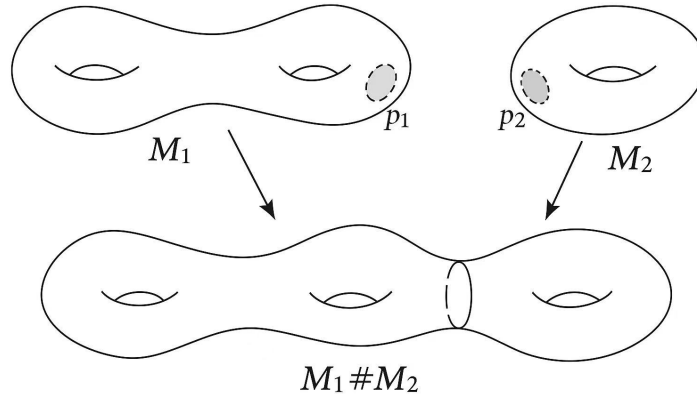


Figure 3: An illustration of the connected sum construction (Alexandrov, 2008).

Definition 2.19 (Lie group). G is a *Lie group* if G is a smooth manifold and a group, where the group operation $\mu : G \times G \rightarrow G$ such that $\mu(g, h) = gh$ and inversion $i : G \rightarrow G$ such that $i(g) = g^{-1}$ are both smooth maps.

The action of a Lie group G on a smooth manifold M is a smooth map $\cdot : G \times M \rightarrow M$ satisfying the group action conditions.

Example 2.20. $\mathrm{GL}(n, \mathbb{R})$ is indeed a Lie group under multiplication. Similarly, many matrix groups are Lie groups, e.g. $\mathrm{SL}_n, \mathrm{SO}_n, \mathrm{O}_n, \mathrm{SU}_n, \mathrm{U}_n$, etc.

Example 2.21. The conformal group $\mathrm{Conf}(\mathbb{R}^n)$ on $\mathbb{R}^n$ consists of smooth transformations that preserve angles. For $n \geq 3$, it is a Lie group isomorphic to:

$$\mathrm{Conf}(\mathbb{R}^n) \cong \mathrm{O}(n+1, 1)/\{\pm 1\},$$

of dimension $\frac{(n+1)(n+2)}{2}$. It is generated by translations, rotations, scalings, and inversions. Here, $O(n+1, 1)$ is the indefinite orthogonal group, consisting of all linear transformations that preserve a bilinear form of signature $(n+1, 1)$.

Example 2.22. The Lie group $SL(2, \mathbb{R})$ acts smoothly on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$ by Möbius transformations:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z \mapsto \frac{az + b}{cz + d} \quad \text{where} \quad ad - bc = 1.$$

2.2 Tangent spaces

The whole point of a tangent space is to describe the “movement” or “direction” at a point, providing a framework for differentiation and vector fields. Moreover while theorems like the hairy ball theorem can be intuitively stated, tangent spaces allow us to state them more formally.

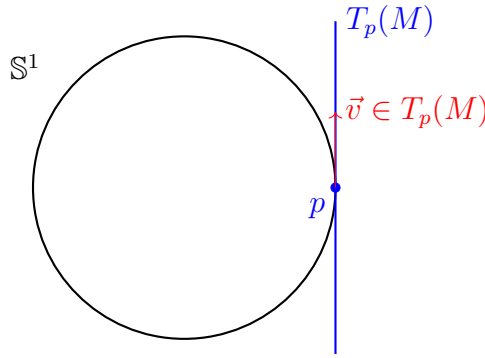


Figure 4: Diagram of the tangent space of $\mathbb{S}^1$ at a point p (OpenAI ChatGPT, 2025)

2.2.1 Curves approach

The first idea to define tangent vectors at a point $p \in M$ is to regard tangent vectors at p as tangents to curves through p . In fact, one could think of $\mathbb{S}^2 \subset \mathbb{R}^3$ for a geometric picture.

Definition 2.23 (Curve on a manifold). Fix $\epsilon > 0$, a *curve* α on a manifold M is a smooth map $\alpha : (-\epsilon, \epsilon) \rightarrow M$. We abbreviate $I = (-\epsilon, \epsilon)$.

If we think of the example of $\mathbb{S}^2$, there are many curves passing through $(1, 0, 0)$ with tangent $(0, 0, 1)$. Therefore, we must take a quotient of “curves with the same tangent”.

Definition 2.24. We say curves $\alpha_1, \alpha_2 : I \rightarrow M$ are *tangent* at $p \in M$ if

1. $\alpha_1(0) = \alpha_2(0) = p$
2. There exist a chart $\phi : U \rightarrow V$ where $p \in U$ such that

$$\left. \frac{d(\phi \circ \alpha_1)(t)}{dt} \right|_{t=0} = \left. \frac{d(\phi \circ \alpha_2)(t)}{dt} \right|_{t=0}$$

Check: Note that the definition of curves being tangent is independent of charts: if α_1, α_2 are tangent with respect to $\phi : U_1 \rightarrow V_1$, and $\psi : U_2 \rightarrow V_2$ is another chart containing p , then

$$\left. \frac{d(\psi \circ \alpha_1)}{dt} \right|_{t=0} = \frac{d(\psi \circ \phi^{-1} \circ \phi \circ \alpha_1)}{dt} = (\psi \circ \phi^{-1})'(\phi(p)) \left. \frac{d(\phi \circ \alpha_1)}{dt} \right|_{t=0} \quad (1)$$

and similarly we obtain

$$\left. \frac{d(\psi \circ \alpha_2)}{dt} \right|_{t=0} = \frac{d(\psi \circ \phi^{-1} \circ \phi \circ \alpha_2)}{dt} = (\psi \circ \phi^{-1})'(\phi(p)) \left. \frac{d(\phi \circ \alpha_2)}{dt} \right|_{t=0} \quad (2)$$

By α_1, α_2 tangent at p with respect to $\phi : U_1 \rightarrow V_1$, (1) and (2) are equal. Hence two curves being tangent is a notion independent of charts (intrinsic to the manifold). We can safely denote

$$\alpha'(0) = \left. \frac{d(\psi \circ \alpha)}{dt} \right|_{t=0}$$

and we define two curves $\alpha_1, \alpha_2 : I \rightarrow M$ to be equivalent, denoted $\alpha_1 \sim \alpha_2$, if $\alpha_1'(0) = \alpha_2'(0)$. By the discussion above, if $\alpha_1 \sim \alpha_2$ and $\alpha_2 \sim \alpha_3$, we get $\alpha_1'(0) = \alpha_2'(0) = \alpha_3'(0)$. Reflexivity and symmetry are obvious, therefore $\sim$ is an equivalence relation on the set

$$\mathcal{C}_p = \{\alpha : I \rightarrow M : \alpha(0) = p\}$$

For simplicity, we identify each equivalence class $[\gamma] \in \mathcal{C}_p/\sim$ with its representative γ . Finally, we define the tangent space.

Definition 2.25 (Tangent space). Let M be a smooth manifold. For all $\gamma \in \mathcal{C}_p/\sim$, $\gamma'(0)$ is called the *tangent vector* of M at $p \in M$. The *tangent space* of M at p , denoted $T_p M$, is the set of all tangent vectors at p .

Definition 2.26 (Tangent bundle). The *tangent bundle* TM of a smooth manifold M as a set is the disjoint union of all tangent spaces of M :

$$TM := \coprod_{p \in M} T_p M$$

The charts of M provides a smooth structure on TM , which will be separately discussed later.

Remark 2.27. A tangent vector $v = [\alpha] \in T_p M$ acts on functions on M by

$$v(f) = \left. \frac{d(f \circ \alpha)}{dt} \right|_{t=0}$$

where α is any representative of the equivalence class v , and the result $v(f)$ is independent of the choice of a representative, by definition of curves being tangent at p . This relates tangent vectors at p to *directional derivatives* at p .

2.2.2 Derivational approach

The derivation is an intrinsic way of generalizing directional derivatives. Recall that for $U \subset \mathbb{R}^n$ open, the directional derivative of a smooth map $f : U \rightarrow \mathbb{R}$ at $p \in U$ in direction of $v \in \mathbb{R}^n$ is defined by

$$v(f) = \lim_{t \rightarrow 0} \frac{f(p + tv) - f(p)}{t}$$

We recall from Analysis II that:

$$v(f) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}$$

which reinforces Remark 1.18. To summarise, tangent vectors are

1. Directions on a manifold, which can be added and multiplied by scalars, forming a vector space
2. They can act on functions and output directional derivatives. Moreover, the action should also be linear: the directional derivative along $v + w$ should equal the sum of the directional derivatives along v and w .

In the next subsection, we will show the bijection between

$$\{D_v : v \in C_p/\sim\} \leftrightarrow \{\text{linear maps } T : C^\infty(M) \rightarrow \mathbb{R}\}$$

where $v = [\alpha]$ and $D_v(f) = \frac{d(f \circ \alpha)}{dt}$, and $C^\infty(M)$ denotes the set of all smooth real-valued maps on M .

The discussion motivates the following algebraic definition of a tangent vector.

Definition 2.28 (Derivation at $p \in M$). A map $D : C^\infty(M) \rightarrow \mathbb{R}$ is a *derivation at $p \in M$* if D is $\mathbb{R}$ -linear and satisfies the Leibniz rule: for all $f, g \in C^\infty(M)$

$$D(fg) = f(p)Dg + g(p)Df$$

A derivation at $p \in M$ is also called a *tangent vector* at p .

Definition 2.29 (Tangent space of M at p). $D_p M$ is the set of all derivations at p . Note that this is an $\mathbb{R}$ -vector space:

$$\begin{aligned} (D_1 + D_2)(f) &= D_1 f + D_2 f \\ (\alpha D_1)f &= \alpha(D_1 f) \end{aligned}$$

for all $D_1, D_2 \in T_p M$, $f \in C^\infty(M)$ and $\alpha \in \mathbb{R}$.

Proposition 2.30 (Properties of derivations). *Let $p \in M, v \in T_p(M), f, g \in C^\infty(M)$:*

1. *If f is a constant function, then $v(f) = 0$.*
2. *If $f(a) = g(a) = 0$, then $v(fg) = 0$.*

Proof. 2 follows immediately from the Leibniz identity of derivations. For 1 note that

$$v(1) = v(1 \cdot 1) = v(1) + v(1) = 2v(1)$$

So $v(1) = 0$, and by linearity $v(c) = 0$ for any constant c .

2.2.3 Equivalence of definitions

In this section, we show that the curves approach and derivational approach to tangent vectors are equivalent. To be precise we show that:

Proposition 2.31. *For $p \in M$, fix an open set U containing p . There exist a bijective mapping $\mathcal{A} : \mathcal{C}_p/\sim \rightarrow D_p$ given by taking $v = [\alpha]$ to the derivation $D_v(f) = (f \circ \alpha)'(0)$ for $f \in C^\infty(U)$.*

Proof. The map is well defined: any two representatives of the equivalence class $[\alpha]$ are tangent. We first prove injectivity: Suppose $v = [\alpha], w = [\beta]$ such that $D_v = D_w$ and pick a chart (ϕ, U) with $p \in U$. By assumption $D_v([x^i]) = D_w([x^i])$ for $1 \leq i \leq n$, so by linearity of derivations we have $D_v(x^i) = D_w(x^i)$, expand to get $(\phi \circ \alpha)'(0) = (\phi \circ \beta)'(0)$. So $\alpha \sim \beta$ and $v = w$.

For surjectivity, let $D \in D_p$ be arbitrary. Pick a chart (ϕ, U) with $p \in U$. WLOG (by maximality of atlas) we can assume that $\phi(p) = 0$. Let $v_i = D(\phi_i)$. We will state a lemma from Lee, 2002 about the coordinate representation of derivations, and as the proof of this lemma requires quite a bit of technical details, we'll cite it without proof.¹

¹The idea is to show that these vectors spans for $T_p \mathbb{R}^n$, then use the differential.

Lemma 2.32. For $p \in M$ and any chart (U, ϕ) ($\phi = (x^1, x^2, \dots, x^n)$ containing p). The coordinate vectors

$$\frac{\partial}{\partial x^i} : f \mapsto \frac{\partial}{\partial x^i} \Big|_{\phi(p)} (f \circ \phi^{-1}) = \frac{\partial \hat{f}}{\partial x^i}(\hat{p})$$

where $\hat{f} = f \circ \phi^{-1}$, $\hat{p} = \phi(p)$ are coordinate representations of f and p , forms a basis of $T_p M$. In coordinates, for all $D \in T_p M$

$$D = \sum_{i=1}^n D(x^i) \left(\frac{\partial}{\partial x^i} \right)_p = \sum_{i=1}^n v^i \left(\frac{\partial}{\partial x^i} \right)_p$$

By lemma, it remains to construct a curve α such that $\alpha(0) = p$ and $(\phi \circ \alpha)'(0) = (v^1, \dots, v^n)$, as we then have

$$(f \circ \alpha)'(0) = (f \circ \phi^{-1})'((\phi \circ \alpha)(0))(\phi \circ \alpha)'(0) = \sum_{i=1}^n v^i \left(\frac{\partial f}{\partial x^i} \right)_p = D(f)$$

By openness of $\phi(U)$ there exist some $\epsilon > 0$ such that $(tv^1, \dots, tv^n) \in \phi(U), \forall t \in (-\epsilon, \epsilon)$. The curve $\alpha : (-\epsilon, \epsilon) \rightarrow M$ by $\alpha(t) = \phi^{-1}(tv^1, \dots, tv^n)$ satisfies this requirement. $\square$

2.2.4 Differentials

Recall that the derivative of a map $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (differentials) at $p \in \mathbb{R}^n$ is the best linear approximation of f near p . There is no such thing as a linear map between two manifolds, but it make good sense to talk about linear maps between tangent spaces. Differentials allow us to relate different tangent spaces, pushing tangent vectors in one tangent space to another.

Definition 2.33 (Differential/Pushforward). Let M, N be smooth manifolds. Given a smooth map $f : M \rightarrow N$, for all $p \in M$ the *differential* is $df_p : T_p M \rightarrow T_{f(p)} N$ such that:

$$df_p(v)[g] = v(g \circ f)$$

for all $v \in T_p M, g \in C^\infty(N)$.

We check that $df_p(v)$ is indeed a derivation at $f(p)$: it inherits linearity from $v \in T_p M$, and satisfies the Leibniz rule: for $g, h \in C^\infty(N)$

$$df_p(v)[gh] = v((gh) \circ f) \tag{3}$$

$$= v((g \circ f)(h \circ f)) \tag{4}$$

$$= v(g \circ f)h(f(p)) + g(f(p))v(h \circ f) \tag{5}$$

$$= df_p(v)[g]h(f(p)) + g(f(p))df_p(v)[h] \tag{6}$$

Directions on manifolds are made precise using tangent spaces. The differential df_p acts as a machine that tells you: if you are at a point $p \in M$ and move tangentially along a vector $v \in T_p M$, then at the point $f(p) \in N$ you will move in the direction $df_p(v) \in T_{f(p)} N$. Here, tangent vectors are understood via the curve-based definition.

Proposition 2.34. If f is a diffeomorphism then $df_p : T_p M \rightarrow T_{f(p)} N$ is an isomorphism of vector spaces. The inverse $(df_p)^{-1} = d(f^{-1})_{f(p)}$

Proof. Linearity is easy to check. For the inverse, we need to show that $(d(f^{-1})_{f(p)}) \circ (df_p)(v) = v$ i.e. we need to show that LHS act on g is precisely $v(g)$ for any $g \in C^\infty(M)$. For $v \in T_p M$, $df_p(v)$ is a derivation sending $g \mapsto v(g \circ f)$. For $w \in T_{f(p)} N$, $d(f^{-1})_{f(p)}(w)$ is a derivation sending $h \mapsto w(h \circ f^{-1})$. Hence for $g \in C^\infty(M)$ we have

$$(d(f^{-1})_{f(p)}) \circ (df_p)(v)[g] = df(v)(g \circ f^{-1}) = v(g \circ f^{-1} \circ f) = v(g) \quad \square$$

In light of the analogy between differentials on manifolds and their counterparts in multivariable calculus, we now examine the coordinate representation of differentials and reveal that they are, in essence, an intrinsic formulation of the *Jacobian*.

To simplify notations (avoid writing down messy transition functions), we consider the special case where M and N are open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $f : M \rightarrow N$ be a smooth map and $p \in M$. We use the basis given in Prop 2.32 to write down the matrix of df_p : let $E = \{\partial/\partial x^i|_p\}$ be a basis of $T_p M$ and $F = \{\partial/\partial y^j|_{f(p)}\}$ a basis of $T_{f(p)} N$. For any $g \in C^\infty(N)$,

$$df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) g = \frac{\partial}{\partial x^i} \Big|_p (g \circ f) = \frac{\partial g}{\partial y^j}(f(p)) \frac{\partial f^j}{\partial x^i}(p) = \left(\frac{\partial f^j}{\partial x^i}(p) \frac{\partial}{\partial y^j} \Big|_{f(p)} \right) g.$$

Hence

$$df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial f^j}{\partial x^i}(p) \frac{\partial}{\partial y^j} \Big|_{f(p)}$$

Thus, with respect to bases E and F of $T_p M$ and $T_{f(p)} N$ respectively, the matrix representation of the differential is:

$${}_F[df_p]_E = \begin{pmatrix} \frac{\partial f^1}{\partial x^1}(p) & \frac{\partial f^1}{\partial x^2}(p) & \cdots & \frac{\partial f^1}{\partial x^n}(p) \\ \frac{\partial f^2}{\partial x^1}(p) & \frac{\partial f^2}{\partial x^2}(p) & \cdots & \frac{\partial f^2}{\partial x^n}(p) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1}(p) & \frac{\partial f^m}{\partial x^2}(p) & \cdots & \frac{\partial f^m}{\partial x^n}(p) \end{pmatrix}$$

Suppose we have charts $(U, \varphi), (V, \psi)$ on M , and v a tangent vector at p . Let $(\partial/\partial x^i|_p), (\partial/\partial \tilde{x}^i|_p)$ be the standard basis with respect to (U, φ) and (V, ψ) , respectively. It's natural to ask how representations of v with respect to these two bases are related.

We introduce some cleaner notation: the transition map is written in the Euclidean style as

$$\psi \circ \varphi^{-1} = (\tilde{x}^1, \tilde{x}^2, \dots, \tilde{x}^n)$$

By a similar derivation as above, we arrive at

$$\frac{\partial}{\partial x^i} \Big|_p = \frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p}) \frac{\partial}{\partial \tilde{x}^j} \Big|_{\varphi(p)}$$

It remains to apply the map to $v^j \partial/\partial x^i|_p = \tilde{v}^j \partial/\partial \tilde{x}^i|_p$

Proposition 2.35 (Change of variables formula).

$$\tilde{v}^j = \frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p}) v^i$$

2.2.5 Tangent bundle

We recall from Definition 2.26 that the *tangent bundle* of M is a disjoint union of tangent spaces over all points on M :

$$TM := \coprod_{p \in M} T_p M$$

This disjoint union is equivalent to the set of ordered pairs (p, v) , where $p \in M, v \in T_p M$. We remark that there is a natural projection map $\pi : TM \rightarrow M$ by $\pi(v, p) = p$ ².

Proposition 2.36 (Smooth structure on tangent bundles). *Let M be an n -manifold, there exist a smooth structure on TM such that TM is a $2n$ -manifold. And $\pi : TM \rightarrow M$ is smooth.*

Proof. Our first step is to define charts for TM . For any chart (U, ϕ) on M , the subset $\pi^{-1}(U) \subset TM$ is the tangent spaces of points in U . By Lemma 2.32 on coordinate representation of tangent vectors, we could define $\tilde{\varphi} : \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$ by

$$\tilde{\varphi} \left(\sum_{i=1}^n v^i \frac{\partial}{\partial x_i} \Big|_p \right) = (\varphi(p), v)$$

taking a derivation at p to p and its coordinates under the standard basis of $T_p M$. The image is clearly $\varphi(U) \times \mathbb{R}^n$.

For charts $(U, \varphi), (V, \psi)$ on M , the corresponding charts we put on TM are going to be $(\pi^{-1}(U), \tilde{\varphi})$ and $(\pi^{-1}(V), \tilde{\psi})$. Let's check compatibility: applying the change of variables formula,

$$\tilde{\psi} \circ \tilde{\varphi}^{-1} (x^1, \dots, x^n, v^1, \dots, v^n) = \left(\psi \circ \phi^{-1}(x), \frac{\partial \psi \circ \phi^{-1}}{\partial x^j}(x) v^j, \dots, \frac{\partial \psi \circ \phi^{-1}}{\partial x^j}(x) v^j \right).$$

which is a smooth map.

By choosing a countable cover $\{U_i\}$ of M consisting of smooth coordinate domains, we obtain a corresponding countable cover of TM by coordinate domains $\{\pi^{-1}(U_i)\}$, which satisfy the conditions of the so called smooth manifold chart lemma (Lee, 2002).

To verify the Hausdorff condition (v), observe that any two points lying in the same fiber of π are contained in a single chart. If instead (p, v) and (q, w) lie in distinct fibers, then there exist disjoint smooth coordinate neighborhoods U and V in M such that $p \in U, q \in V$, and $\pi^{-1}(U)$ and $\pi^{-1}(V)$ form disjoint coordinate neighborhoods of (p, v) and (q, w) , respectively.

Lastly, $\pi : TM \rightarrow M$ is smooth because for chart (U, φ) of M and $(\pi^{-1}(U), \tilde{\varphi})$ of TM ,

$$\varphi \circ \pi \circ \tilde{\varphi}^{-1}(x, v) = \varphi \circ \pi \left(v^i \frac{\partial}{\partial x_i} \Big|_{\phi^{-1}(x)} \right) = \varphi(\varphi^{-1}(x)) = x$$

which is smooth. □

The following examples illustrate topological properties of tangent bundle:

Definition 2.37. A manifold of dimension n is *parallelisable* if $TM \cong M \times \mathbb{R}^n$.

²Technically speaking, as the tangent bundle is a vector bundle over M , π has to satisfy additional properties of bundle projections.

Example 2.38. Vector spaces as manifolds are obviously parallelisable. $\mathbb{S}^1$ is also parallelisable, but $\mathbb{S}^2$ is not.

Example 2.39. Lie groups are parallelisable. Given Lie group G , it acts on itself by left multiplication, inducing an action $d\phi$ on TG . Thus we have an isomorphism $\phi : TG \rightarrow G \times T_e G$ such that for $v \in T_g G$:

$$\phi(v) := (g, d\phi_{g^{-1}}v)$$

2.3 Vector fields and Lie algebras

Definition 2.40 (Vector field). A *vector field* on M is a smooth function $X : M \rightarrow TM$ such that $X(p) := X_p \in T_p M$.

Remark 2.41. Vector fields over M form a vector space $\mathcal{X}(M)$.

Remark 2.42. Vector fields on M are linear functions from $C^\infty(M)$ to $C^\infty(M)$: for any $f \in C^\infty(M)$ and $p \in M$,

$$(Xf)(p) := X_p(f) \in \mathbb{R}$$

Moreover vector fields satisfy the Leibniz rule, $X(fg) = fXg + gXf$. These properties are similar to those of a derivation at a point, but since the above properties hold for all $p \in M$, we just call vector fields *derivations*.

Definition 2.43 (Flow). A *flow* ϕ_X of a vector field X is the solution to the differential equation $\dot{x} = X(x)$, in other words:

$$X = d\phi_X$$

Theorem 2.44. For a compact manifold M , the flow ϕ_X globally exists, i.e. the domain is $\mathbb{R} \times M$. This directly follows from Picard-Lindelöf theorem and smoothness of X .

Now, there is an obvious Lie group action on a compact manifold:

Theorem 2.45. For X a vector field on compact manifold M , there is a Lie group action $\phi_X : (\mathbb{R}, +) \times M \rightarrow M$.

We wonder if we can define a Lie group action using two or more vector fields. However, this is not easy: the set of diffeomorphisms $\phi_{aX+bY}(t, \cdot)$ is not always closed. One issue that arises is that first moving along the flow of X in unit time, then moving along the flow of Y , then $-X$ and $-Y$, we might not end up at our initial position.

The Lie bracket is defined to measure this failure as a vector field. We will see the relation at the end of this section.

Definition 2.46 (Lie bracket). Given two vector fields X, Y on the manifold M , their *Lie bracket* is defined to be:

$$[X, Y](f) = X(Y(f)) - Y(X(f))$$

This is a derivation, while XY generally fails the Leibniz's rule.

Proposition 2.47. Obviously for vector fields:

- $[\cdot, \cdot]$ is bilinear
- $[X, Y] = -[Y, X]$
- $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$

The following definition is an abstraction of space of vector fields:

Definition 2.48 (Lie algebra). A Lie algebra is a vector space equipped with a bilinear form $[\cdot, \cdot]$ satisfying:

$$[a, b] = -[b, a]$$

and

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0$$

Definition 2.49 (Left invariant vector field). A *left invariant* vector field X on Lie group G must satisfy:

$$d(L_g)X = X \circ L_g$$

where

$$L_g : G \rightarrow G, L_g(x) := gx$$

Pretty straight forward: you translate the space, and the vector fields don't change.

Definition 2.50 (Lie algebra of a Lie group). The Lie algebra $\mathfrak{g}$ of a Lie group G is the set of all left invariant vector fields of G ; $\mathfrak{g}$ is isomorphic to $T_e G$.

Proof. We prove $\mathfrak{g} \cong T_e G$. The linear transformation $\varphi : \mathfrak{g} \rightarrow T_e G$ is obviously $\varphi(X) := X_e$. The inverse map $\vartheta : T_e G \rightarrow \mathfrak{g}$ is also obvious, by definition of left invariant vector field: since

$$X_g = (X \circ L_g)(e) = (d(L_g) \circ X)(e)$$

we have

$$\vartheta(\vec{v}) = d(L_g)(\vec{v})$$

□

Definition 2.51 (Exponential map). The *exponential map* $\exp : \mathfrak{g} \rightarrow G$ is $\exp(X) := \phi_X(1, e)$.

And finally, we see what was promised:

Theorem 2.52 (Campbell-Baker-Hausdorff formula). *For elements X, Y in the Lie algebra of a Lie group:*

$$\exp(X) \times \exp(Y) = \exp\left(X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}([X, [X, Y]] + [Y, [Y, X]]) + \dots\right)$$

All terms are represented in X, Y using Lie brackets.

Theorem 2.53. *A finite-dimensional Lie algebra $\mathfrak{g}$ over $\mathbb{R}$ or $\mathbb{C}$ attains a unique simply-connected Lie group G whose Lie algebra is $\mathfrak{g}$.*

And:

Proposition 2.54. *For a compact manifold M , a finite dimensional subspace of $\mathcal{X}(M)$ which is closed under Lie bracket defines a finite dimensional Lie algebra and thus a simply connected Lie group acting on it.*

We are not going to prove or use these theorems. The reason they are here is to provide a complete reasoning for defining Lie bracket and Lie algebra.

2.4 Supplementary theorems

Definition 2.55 (Submersion). A smooth map $f : M \rightarrow N$ is called a *submersion* if:

$$\forall p \in M, df_p : T_p M \rightarrow T_{f(p)} N \text{ is surjective}$$

Theorem 2.56 (Submersion theorem). *For $f : M \rightarrow N$ a submersion, exists a chart (ϕ, U) of M and a chart (ψ, V) of N such that:*

$$\psi \circ f \circ \phi^{-1}$$

is an orthogonal projection $\mathbb{R}^{\dim M} \rightarrow \mathbb{R}^{\dim N}$, simply by removing the last $\dim M - \dim N$ coordinates of the vector.

Submersion theorem is a consequence of inverse function theorem.

Theorem 2.57 (Closed subgroup theorem). *(Lee, 2002 Theorem 20.12) G is a Lie group, K is a closed subgroup, then K is an embedded submanifold.*

Theorem 2.58 (Quotient Manifold theorem). *(Lee, 2002 Theorem 21.10) M is a manifold, G is a Lie group smoothly acting on M . The space of orbits obtains a unique smooth structure, coinciding with the quotient topology.*

These two theorems are very subtle to show.

3 Homogeneous spaces

3.1 Definition

Definition 3.1 (Isotropy group). For a group action G on M , the isotropy group (or stabiliser) at a point $p \in M$ is the subgroup $G_p = \text{Stab}(p)$ of G consisting of group elements which fix p . In other words, for all $g \in G_p$, we have that $g \cdot p = p$.

Definition 3.2 (Homogeneous space). A manifold is called homogeneous if there exists a Lie group transitively and smoothly acting on it.

We examine the relation between the space, the Lie group, and the isotropy subgroup.

Proposition 3.3. *The isotropy group G_p is always closed in G , since the group action is continuous, and the preimage of the singleton $\{p\}$ under the action is closed.*

Proposition 3.4. *For a transitive Lie group action $G \times M \rightarrow M$, take $p \in M$, the following map is a submersion:*

$$\tau : G \rightarrow M, \tau(g) := g \cdot p$$

Theorem 3.5 (Homogeneous representation theorem). *For a transitive Lie group action $G \times M \rightarrow M$:*

$$\forall p \in M, M \cong G/G_p$$

where by closed subgroup theorem, G_p is an embedded manifold, properly acting on G by right multiplication, with orbits as left coset of G_p , providing a smooth structure on G/G_p (quotient manifold theorem). G/G_p is called a homogeneous representation of M .

3.2 Examples of homogeneous spaces

Example 3.6. Obviously, Lie groups themselves are homogeneous spaces, this includes $\mathbb{R}^n$, Matrix groups, etc.

Example 3.7. Given Lie groups $H \subseteq G$, coset manifold G/H is automatically a homogeneous space by left multiplication action.

Example 3.8 (n -sphere). $\mathbb{S}^n$ are homogeneous space with $\text{SO}(n+1)$ transitively acting on it. The isotropy subgroup of $e_1 = (1, 0, \dots, 0) \in \mathbb{S}^n$ is the rotational subgroup that leaves the plane normal to e_1 invariant. Concretely it's the set of matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$$

where $B \in \text{SO}(n)$, hence $\mathbb{S}^n \cong \text{SO}(n+1)/\text{SO}(n)$.

Example 3.9. $\text{SL}(2, \mathbb{R})$ acts on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$ such that each matrix corresponds to a Möbius transformation:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}$$

For all $z', w' \in \mathbb{H}$, from complex analysis we know that we can construct a Möbius transformation

$$f(z) = \frac{az + b}{cz + d}$$

such that $ad - bc = 1$ and $f(z') = w'$, so $\mathrm{SL}(2, \mathbb{R})$ is a transitive action. This is also smooth so $\mathbb{H}$ is homogeneous. Moreover, it is simple to find that $\mathrm{Stab}(i) = \mathrm{SO}(2)$ so $\mathbb{H}$ is diffeomorphic to $\mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2)$.

Example 3.10 (Product space). For two homogeneous space M_1, M_2 with Lie group action by G_1, G_2 respectively, their product $M_1 \times M_2$ attains a transitive group action:

$$(g_1, g_2) \cdot (m_1, m_2) := (g_1 \cdot m_1, g_2 \cdot m_2)$$

The isotropy group is:

$$(G_1 \times G_2)_{(m_1, m_2)} = (G_1)_{m_1} \times (G_2)_{m_2}$$

This makes the torus a homogeneous space.

Example 3.11 (Grassmann manifold). The *Grassmannian* $G_k(V)$ is the set of all k -dimensional vector subspaces of an n -dimensional vector space V . $\mathrm{GL}(V)$ acts on $G_k(V)$ via left multiplication, and this action is transitive as any two k -dimensional subspaces are linearly isomorphic. Fix $U \in G_k(V)$: the isotropy subgroup of U can be identified with the isotropy subgroup of $\mathbb{R}^k$ in $\mathrm{GL}(n, \mathbb{R})$, which is the set of block upper-triangular matrices of the form:

$$D = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$$

where $A \in \mathrm{GL}(k, \mathbb{R})$, $C \in \mathrm{GL}(n - k, \mathbb{R})$ and $B \in M(k \times (n - k), \mathbb{R})$. The isotropy subgroup of U is accordingly isomorphic to:

$$\mathrm{GL}(k, \mathbb{R}) \times \mathrm{GL}(n - k, \mathbb{R}) \times M(k \times (n - k), \mathbb{R})$$

Grassmannians are clearly a generalisation of real projective spaces; we can go one step further and generalise Grassmannians to real flag manifolds.

Example 3.12 (Real flag manifolds). Let V be an n -dimensional vector space and $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{Z}_{\geq 0}^m$ such that $\alpha_1 < \dots < \alpha_m < n$. A *real flag manifold* of type α , denoted $F_\alpha(V)$, is the set of all *flags*, nested sequences of vector subspaces $U_1 \leq \dots \leq U_m \leq V$ such that $\dim U_i = \alpha_i$. We can define a $\mathrm{GL}(V)$ action on $F_\alpha(V)$ such that

$$T \cdot (U_1, \dots, U_m) = (T(U_1), \dots, T(U_m))$$

This is a transitive action: given two flags $(U_1, \dots, U_m), (V_1, \dots, V_m)$ of type α , write bases $u_1, \dots, u_n$ and $v_1, \dots, v_n$ of V such that $u_1, \dots, u_{\alpha_k}$ is a basis of U_k and $v_1, \dots, v_{\alpha_k}$ is a basis of V_k , then define the basis isomorphism $T : u_k \mapsto v_k$, and we have $T \cdot (U_1, \dots, U_m) = (V_1, \dots, V_m)$.

In the case that $\alpha = (1, 2, \dots, m)$, $F_\alpha(V)$ is a *complete flag manifold*, and its isotropy subgroups are isomorphic to B_n , the group of $n \times n$ upper-triangular matrices. Hence $F_\alpha(V) \cong \mathrm{GL}(V)/B_n$.

Example 3.13 (Klein bottle). The Klein bottle K is diffeomorphic to the torus $\mathbb{T}^2$ with antipodal points identified: in other words, $K \cong \mathbb{T}^2/\mathbb{Z}_2$. $\mathbb{R}^2$ acts on the torus by translation so it acts on K too.

Example 3.14 (Trefoil knot complement in $\mathbb{S}^3$). Let K denote the trefoil knot in $\mathbb{S}^3$. The knot complement $\mathbb{S}^3 \setminus K$ is diffeomorphic to $\mathrm{SL}(2, \mathbb{R})/\mathrm{SL}(2, \mathbb{Z})$ (Liu, 2012).

Example 3.15 (Stiefel manifolds). An *orthonormal k -frame* is a tuple of k orthonormal vectors in $\mathbb{R}^n$. The set of all orthonormal k -frames in $\mathbb{R}^n$ is called the *Stiefel manifold* $V_k(\mathbb{R}^n)$. We

can define an $\mathrm{SL}(n, \mathbb{R})$ action on $V_k(\mathbb{R}^n)$. Given $A \in \mathrm{SL}(n, \mathbb{R})$ and an orthonormal k -frame $(v_1, \dots, v_k)$, the action is defined:

$$A \cdot (v_1, \dots, v_k) = (Av_1, \dots, Av_k)$$

This action is transitive as given any two orthonormal k -frames, we can extend them to orthonormal $\mathbb{R}^n$ bases of the same orientation via Gram-Schmidt. The change of basis matrix is clearly orthogonal and has determinant 1 since the bases have the same orientation, hence it is in $\mathrm{SL}(n, \mathbb{R})$.

3.3 Structure of homogeneous spaces and non-homogeneous spaces

We've seen so many homogeneous spaces. We naturally raise a question, what are the non-homogeneous spaces? It's hard to enumerate all the Lie groups and come to a conclusion, so we want to see what are the properties of homogeneous spaces.

Example 3.16. $\mathbb{T}^2 \amalg \mathbb{S}^2$, the disjoint union of a torus and a sphere, is non-homogeneous. There is no homeomorphism mapping the torus to the sphere, thus no transitive group action on this space.

This example is too trivial so we would like to look for connected non-examples. To this end, we will discuss algebraic topology properties for homogeneous spaces.

3.3.1 Fundamental groups of homogeneous spaces with compact Lie groups

In this section, we will conclude that for a compact Lie group G and a closed subgroup H , taking any two loops in G/H , we can repeat them for some times and make them commutative.

Definition 3.17 (Homotopy of paths). In a topological space X , given two path $\alpha, \beta : [0, 1] \rightarrow X$ with same endpoints $\alpha(0) = \beta(0), \alpha(1) = \beta(1)$. They are homotopic iff

$$\exists H : [0, 1]^2 \rightarrow X, H(0, t) = \alpha(t), H(1, t) = \beta, H(t, 0) = \alpha(0), H(t, 1) = \alpha(1)$$

This is obviously an equivalence relation among all paths.

Definition 3.18 (Joining two paths). Given $\gamma : [0, 1] \rightarrow X, \theta : [0, 1] \rightarrow X$ both continuous, and $\gamma(1) = \theta(0)$, then they can be joined together into a new path:

$$(\gamma \cdot \theta)(x) := \begin{cases} \gamma(2x) & x \in [0, 1/2] \\ \theta(2x - 1) & x \in [1/2, 1] \end{cases}$$

Definition 3.19 (Fundamental Group). The fundamental group of a topological space based at a point $p \in X$ consists of homotopy classes of loops $\gamma : [0, 1] \rightarrow X, \gamma(0) = \gamma(1) = p$. The group operation is well defined:

$$[\alpha] \cdot [\beta] := [\alpha \cdot \beta]$$

Obviously this operation does satisfy all group axioms - the inversion of a loop is $\gamma^{-1}(t) := \gamma(1 - t)$.

The group is denoted by $\pi_1(X, p)$. Note that a path connected space has fundamental group isomorphic at all points, with isomorphism determined by a path. Thus sometimes we can use $\pi_1(X)$ for simplification.

Example 3.20. Klein bottle has fundamental group $\langle a, b \mid ab = b^{-1}a \rangle$. Note $ab^{-1} = b(b^{-1}a)b^{-1} = b(ab)b^{-1} = ba$. For an element of this group, we obtain a canonical representation by moving all b terms to the beginning of the representation, e.g.:

$$ba^3b^2 = baaabb = baab^{-1}ab = babaab = bb^{-1}aaab = abaa = b^{-1}a^3$$

Note the rule is $a^mb^n = b^{n \cdot (-1)^m}a^m$ which can be proved by induction. So the group elements are of the form b^na^m . Now we claim $g^2h^2 = h^2g^2$ for all elements of the fundamental group:

$$(b^na^m)^2 = b^n(a^mb^n)a^m = \begin{cases} b^{2n}a^{2m} & m \text{ even} \\ a^{2m} & m \text{ odd} \end{cases}$$

So g^2 is always in the form of $b^{2n}a^{2m}$. Thus $g^2h^2 = b^{2n}a^{2m}b^{2k}a^{2l} = b^{2n+2k}a^{2m+2l} = h^2g^2$

Example 3.21. Double torus doesn't have the above property - The circulations along the two exterior holes never commute, no matter how many times were repeated. So with the statement in the beginning of this subsection, double torus cannot be represented as G/H where G is a compact Lie group and H is a closed Lie subgroup.

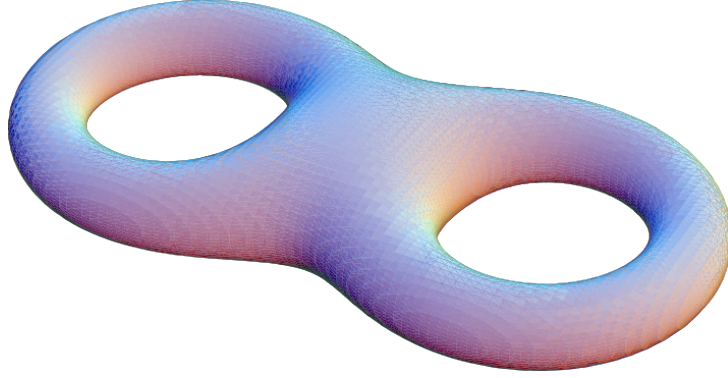


Figure 5: A double torus. Made with Wolfram Mathematica.

Lemma 3.22. A path γ in G/H can be lifted to a path $\tilde{\gamma}$ in G such that $\gamma = \pi \circ \tilde{\gamma}$ where $\pi : G \rightarrow G/H$ is the projection map.

Proof. We want $\tilde{\gamma} : I \rightarrow G$. An argument is to prove that the maximal interval I_{max} such that $\exists \theta : I_{max} \rightarrow G, \gamma = \pi \circ \theta$, I_{max} must be non-empty, closed, and open - makes $I_{max} = I$. Obviously due to continuity I_{max} is closed, and $0 \in I_{max}$.

For $x \in I_{max}$, want to prove that there is an open ball where we can continue this path θ . We reduce to the case that $\theta(x) = e_G$ by left translation.

Now, $f : G \rightarrow G/H$ being a submersion. Using submersion theorem, the path can be locally lifted to G .

Thus, I_{max} is an open, closed, and non-empty subset of I . So $I_{max} = I$.

□

Remark 3.23. The word “lift” is because when H is discrete, G is a covering space of G/H and the theorem reduces to path lifting property.

Theorem 3.24. The fundamental group of a connected Lie group is abelian.

Proof. For $\gamma : [0, 1] \rightarrow G$, $\theta : [0, 1] \rightarrow G$ loops, we give the following homotopy:

$$H(t, x) := \begin{cases} \gamma(2x) & 2x \leq t \\ \gamma(t) \times \theta(2x - t) & t \leq 2x \leq 1 + t \\ \gamma(2x - 1) & 2x \geq 1 + t \end{cases}$$

Where $\times$ is the group multiplication. This proves:

$$\gamma \cdot \theta \text{ homotopic to } \theta \cdot \gamma$$

□

Remark 3.25. The above theorem and proof also work for path-connected topological groups.

Theorem 3.26. *For a compact Lie group G , and a Lie subgroup H , the fundamental group of G/H has the following property:*

$$\{g^n \mid g \in \pi_1(G/H)\} \text{ is abelian}$$

Where $n < \infty$ is the number of connected components of H .

Proof. First of all, since G is compact, H is closed subset of compact space, thus compact. So H only has finitely many connected components. Since H is a manifold, connected components are path components, which can be proved using local Euclidean property.

Now consider a loop in G/H , lifted to a path γ in G . By right multiplying an element of H we can let γ start at any point in H .

Now, $\gamma_1 := \gamma$, $\gamma_{m+1} := \gamma_m \cdot (\gamma \times \gamma_m(1))$ Where $\cdot$ denotes the connection of two paths. $\gamma_m(1) = \gamma(1)^m$. Considering the finite quotient group $H/C_e H$ where $C_e H$ is the connected component of H containing e , by topology is normal subgroup. $|H/C_e H| = n$ thus $\gamma(1)^n C_e H = e C_e H$, Thus $\gamma_n(1)$ is in the same path component of H as e .

Now take a path δ inside H connecting the two group elements, $\gamma_n \cdot \delta$ gives a loop in G , and $\pi \circ (\gamma_n \cdot \delta)$ homotopic to the original path taken in G/H repeated for n times.

So a path in G/H , repeated for n times gives a loop in G , while $\pi_1(G)$ is abelian. □

Remark 3.27. Example 3.14 gives a counter-example to this theorem for H with infinite pieces - $\pi_1 \left(\frac{\text{SL}(2, \mathbb{R})}{\text{SL}(2, \mathbb{Z})} \right) \cong \langle a, b \mid a^2 = b^3 \rangle$. However, the theorem stands for even non-compact homogeneous spaces as long as H has finitely many path components.

3.3.2 Non-example of a homogeneous space

The following result comes from Mostow, 2005.

Theorem 3.28. *The Euler characteristic of a compact homogeneous space is non-negative.*

We are not formally defining Euler characteristic using singular homology since the definition is too long. The definition is pretty standard.

Remark 3.29. We have $\chi = V - E + F$ for polyhedron surfaces which are partitioned into simplices. For spaces of higher dimensions:

$$\chi = V_0 - V_1 + V_2 - V_3 + V_4 - \dots$$

$E = V_1$, $F = V_2$ and V_n denotes the number of n simplices.

Proposition 3.30. (*Basic properties of Euler characteristic*)

- $\chi(\mathbb{S}^n) = 1 + (-1)^n$
- $\chi(M \times N) = \chi(M)\chi(N)$
- $\chi(M \# N) = \chi(M) + \chi(N) - \chi(\mathbb{S}^n)$

Example 3.31. So a double torus $X := (\mathbb{S}^1 \times \mathbb{S}^1) \# (\mathbb{S}^1 \times \mathbb{S}^1)$ has Euler characteristic -2 and can't be a homogeneous space.

Remark 3.32. The group of all diffeomorphisms $X \rightarrow X$ where X is the double torus, transitively acting on X . However, it can't be a manifold, thus not a Lie group.

Example 3.33. $\chi((\mathbb{S}^2 \times \mathbb{S}^2) \# (\mathbb{S}^1 \times \mathbb{S}^1 \times B^2) \# (\mathbb{S}^1 \times \mathbb{S}^1 \times B^2)) = 0$, but the fundamental group argument can show that it cannot be transitively acted by a compact Lie group.

4 Symmetric spaces

A large class of homogeneous spaces are *symmetric spaces*. These spaces have special geometric properties which can be built up from many different angles. We will focus on constructing the properties from symmetric spaces out of *geodesics*.

4.1 Riemannian manifolds

To build up a symmetric space, we need the notion of an isometry, which is a distance-preserving map on a manifold. On a typical smooth manifold however, we do not *a priori* have any notion of “distance”. We hence need to introduce a metric onto our manifold.

Definition 4.1 (Riemannian metric). Let M be a smooth manifold. A *Riemannian metric* on M assigns to each $p \in M$ a positive-definite, symmetric, bilinear form $\langle \cdot, \cdot \rangle_p$ on the tangent space $T_p M$ which varies smoothly³: if $\mathbf{x} \in U \subset \mathbb{R}^n \rightarrow M$ is a system of coordinates around p , with $\mathbf{x}(x_1, \dots, x_n) = q \in \mathbf{x}(U)$ and $\frac{\partial}{\partial x_i}(q) = d\mathbf{x}_q(0, 0, \dots, 1, \dots, 0)$, then $\langle \frac{\partial}{\partial x_i}(q), \frac{\partial}{\partial x_j}(q) \rangle_q = g_{ij}(x_1, \dots, x_n)$ is a smooth function on U . A *Riemannian manifold* is a smooth manifold endowed with a Riemannian metric.

Remark 4.2. This induces a norm $\|\cdot\|_p : T_p M \rightarrow \mathbb{R}$ where $\|v\|_p = \sqrt{\langle v, v \rangle_p}$.

Example 4.3 (Euclidean spaces). Let $M = \mathbb{R}^n$ with $\frac{\partial}{\partial x_i} = e_i = (0, \dots, 1, \dots, 0)$. The metric is then given by $\langle e_i, e_j \rangle = \delta_{ij}$, the standard dot product on $\mathbb{R}^n$. This is the (canonical) *Euclidean metric*.

Example 4.4 (Hyperbolic spaces). In contrast, hyperbolic spaces have a Riemannian metric that scales differently depending on location. For example, the upper half space $\{(x_1, \dots, x_n \in \mathbb{R}^n | x_n > 0)\}$ has the Riemannian metric

$$\frac{dx_1^2 + \dots + dx_n^2}{dx_n^2}$$

Remark 4.5. Every smooth manifold admits a Riemannian metric that is not the canonical Euclidean metric (Lee, 2018).

Despite the name, the Riemannian metric above is not an actual metric. However, it can be extended to a metric (with the metric space being M). We define the length of a piecewise smooth curve $\gamma : [0, 1] \rightarrow M$ with $\gamma'(t)$ nonzero everywhere that it is defined as:

$$\text{len}(\gamma) = \int_0^1 \|\gamma'(t)\|_{\gamma(t)} dt$$

Thus a metric can be defined as:

$$d_g(x, y) := \inf\{\text{len}(\gamma) \mid \gamma \text{ is piecewise smooth, } \gamma(0) = x, \gamma(1) = y\}$$

This makes (M, d_g) into a metric space and the induced topology by the metric coincides with the topology on M (Lee, 2018).

³Technically speaking, a Riemannian metric on M is a smooth symmetric 2-tensor field that is positive-definite at all $p \in M$, but this is a technicality and isn't necessary to the following discussion.

4.2 Symmetric spaces

Equipped with this metric, we can now give a proper definition:

Definition 4.6 (Isometry). Let (M, g) be a Riemannian manifold equipped with the metric g . A smooth map $\phi : M \rightarrow M$ is an *isometry* if for all vector fields X, Y on M , we have:

$$g(d\phi_p(X_p), d\phi_p(Y_p)) = g(X_p, Y_p)$$

for all $p \in M$, where:

- X_p, Y_p are tangent vectors at p
- $d\phi_p$ is the differential of ϕ at p

Meaning that distances between vectors are preserved under the map ϕ .

Remark 4.7. The set of all isometries from a Riemannian manifold M to itself forms a group under composition, called the *isometry group*.

We can now begin to build towards a symmetric space:

Definition 4.8 (Point reflections). Let (M, g) be a Riemannian manifold. A *point reflection* at $p \in M$ is an isometry $\sigma : M \rightarrow M$ such that $\sigma(p) = p, d\sigma_p := -\text{Id} : T_p M \rightarrow T_p M$ and $\sigma \circ \sigma = \text{Id}_M$ (the final property is called σ being an *involution*)

This means that our isometry ϕ fixes p , and for any other $v \in M$, $d\phi_p(v) = -v$, in effect acting as multiplication by -1 . Furthermore, applying the isometry twice to the same point returns it to the original position. All in all, near the point, our isometry acts like a reflection about p . We can give a brief example to illustrate this:

Example 4.9. Consider the manifold $\mathbb{R}^2$ with the usual metric, and suppose we have an isometry $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x) = -x + (4, 2)$. It is not difficult to verify that this is an isometry, and that $f(2, 1) = (4, 2) - (2, 1) = (2, 1)$. Furthermore, we have that $df/dx = -\text{Id}$ and thus our isometry is a point reflection about $(2, 1)$.

Finally, we define symmetric spaces formally:

Definition 4.10 (Symmetric spaces). A *(Riemannian) symmetric space* is a (Riemannian) manifold S such that for all $p \in S$, there exists a point reflection at p .

4.3 Properties of symmetric spaces

Symmetric spaces have many results involving group actions that reveal their high degree of symmetry. However, to prove many of these results, we need to first discuss *geodesics*, which informally is a curve representing the locally shortest path between two points on a Riemannian manifold. We now introduce a few concepts that will help us along the way:

Definition 4.11 (Affine connection). Let M be a smooth manifold, and $\Gamma(TM)$ be the space of smooth vector fields on M . An *affine connection* on TM is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TM)$ with $(X, Y) \mapsto \nabla_X Y$, satisfying:

1. $\nabla_{f_1 X_1 + f_2 X_2} Y = f_1 \nabla_{X_1} Y + f_2 \nabla_{X_2} Y$
2. $\nabla_X (a_1 Y_1 + a_2 Y_2) = a_1 \nabla_X (Y_1) + a_2 \nabla_X (Y_2)$ for $a_1, a_2 \in \mathbb{R}$
3. $\nabla_X (fY) = f \nabla_X Y + X(f)Y$

for all $f_1, f_2 \in C^\infty(M)$ and all smooth vector fields $X, Y \in \Gamma(TM)$.

This connection allows us to take derivatives of vector fields along curves:

Definition 4.12. A smooth vector field V along a curve γ is *extendible* if there exists a smooth vector field $\tilde{V}$ on a neighbourhood of the image of γ such that $V(t) = \tilde{V}_{\gamma(t)}$ for each $t \in I$.

Here is a theorem from Lee, 2018:

Theorem 4.13. Let M be a smooth manifold, and ∇ be a connection in TM . For each smooth curve $\gamma : I \rightarrow M$, the connection determines a unique operator $D_t : \Gamma(\gamma) \rightarrow \Gamma(\gamma)$, called the *covariant derivative along γ* such that

1. $D_t(aV + bW) = aD_t(V) + bD_t(W)$ for reals a, b
2. $D_t(fV) = f'V + fD_t(V)$
3. If V is extendible, for each extension $\tilde{V}$, $D_tV(t) = \nabla_{\gamma'(t)}\tilde{V}$

With this in hand, we can finally define geodesics properly:

Definition 4.14 (Geodesics). A *geodesic* is a smooth curve γ such that $D_t\gamma' \equiv 0$. A geodesic is *maximal* if it cannot be extended to a geodesic on a larger interval.

Remark 4.15. Geodesics have many other properties such as uniqueness that we will not cover in this paper. However, the most important result is that a point reflection σ_p takes a geodesic through p and reverses it (do Carmo, 1992, page 73, after corollary 3.9), which we will use in a later proof.

We can now begin to prove that all symmetric spaces are homogeneous:

Definition 4.16 (Geodesic completeness). A Riemannian manifold is *geodesically complete* if every maximal geodesic is defined for all $t \in \mathbb{R}$

In general, Riemannian manifolds are not geodesically complete. However, familiar spaces like $\mathbb{R}^n, \mathbb{S}^n, \mathbb{T}^n$ are geodesically complete.

Example 4.17. The punctured plane $\mathbb{R}^2 \setminus \{0\}$ is not geodesically complete. Any geodesic that passes through the origin cannot be extended past the origin.

Theorem 4.18. *Symmetric spaces are geodesically complete.*

Proof. We take a geodesic γ defined on $[0, s)$ in a symmetric space S , then take $\frac{3s}{4} \in (\frac{s}{2}, s)$. There exists a reflection $\sigma_{\gamma(\frac{3s}{4})}$ since S is a symmetric space. This reflection takes our geodesic γ to the geodesic through $\gamma(\frac{3s}{4})$ with reverse direction and tangent vector $-\frac{d\gamma(t)}{dt}|_{t=\frac{3s}{4}}$. This geodesic coincides with γ on $(\frac{s}{2}, s)$ and extends it to $(s, \frac{3s}{2})$. We can repeat this process indefinitely, thus extending this geodesic to the whole of $\mathbb{R}$. $\square$

Lemma 4.19. *For all $p, q \in S$, there exists $r \in S$ such that $\sigma_r(p) = q$.*

Proof. Since our symmetric space is geodesically complete, by the Hopf-Rinow theorem, there exists a geodesic γ between any two points in the symmetric space. Let r be the midpoint of this geodesic. Then it is clear $\sigma_r(p) = q$ as the reflection preserves distances and reverses the direction of the geodesic. $\square$

Theorem 4.20. *Every symmetric space is homogeneous, with the group being the isometry group on the symmetric space S with composition as the group operation.*

Proof. With the lemma above, the proof is obvious - for any two points on the space there exists an isometry that moves one point to the other, and hence we have a transitive group action. $\square$

In view of this, we can write our symmetric space S as G/K where G is the group of isometries and $K = G_p$ is the isotropy group of a point p under the action G .

Here are some examples of symmetric spaces:

Example 4.21. Let $S = \mathbb{R}^n$ with the Euclidean metric. The symmetry about any point $x \in \mathbb{R}^n$ is the reflection $s_x(x+v) = x-v$. The isometry group is the Euclidean group $E(n)$ generated by translations, rotations, and reflections. About the origin, the isotropy group is the orthogonal group $O(n)$.

Example 4.22. Let $S = \mathbb{S}^n$ with the usual scalar product. The symmetry at any $x \in S$ is the reflection at the line $\mathbb{R}x$, i.e. $s_x(y) = -y + 2\langle y, x \rangle x$. The symmetries generate the full isometry group $O(n+1)$, and the isotropy group of the unit vector e_{n+1} is the group $O(n) \subset O(n+1)$.

4.4 Decomposition of symmetric spaces

We can decompose the Lie algebra of symmetric spaces as such:

Theorem 4.23. *A symmetric space $S \cong G/H$ with Lie algebras $\mathfrak{g} = \text{Lie}(G)$, $\mathfrak{h} = \text{Lie}(H)$ can be decomposed as*

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$$

with $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$, $[\mathfrak{h}, \mathfrak{m}] \subset \mathfrak{m}$, $[\mathfrak{m}, \mathfrak{m}] \subset \mathfrak{h}$.

To prove this result, first we must remark that it is unnecessarily strict to require G be the full isometry group of the symmetric space S . The subgroup $G^0 = I_0(M) \subset I(M) = G$, called the *identity component* of the isometry group, is the largest connected subgroup containing the identity element. Let H be the subgroup of the identity component leaving a point p fixed (i.e. $H = \text{St}_{G^0}(p)$). Then by Theorem 3.3(i) in Chapter IV of Helgason, 1962, $S \cong G^0/H$. From here on we write G^0 as G .

Next, we have that for a symmetry σ_p on S , it gives rise naturally to an involution on G , defined as $\sigma : G \rightarrow G, g \rightarrow \sigma_p g \sigma_p$, called the *Cartan involution*.

Consider the Lie algebras $\mathfrak{g} = \text{Lie}(G)$, $\mathfrak{h} = \text{Lie}(H)$. We note that: a) $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$ and that b) $\mathfrak{h}$ consists of all elements of $\mathfrak{g}$ fixed by σ (Kobayashi and Nomizu, 1969).

Remark 4.24. Recall from linear algebra that if given a real vector space V with $\theta \in \text{End}(V)$ diagonalisable with eigenvalues ± 1 , then $V = \ker(\theta - I) \oplus \ker(\theta + I)$. We can directly apply this result in the case where $\theta^2 = \text{Id}$, as we can deduce that θ has minimal polynomial $x^2 - 1$ and so, since this minimal polynomial has no repeated roots, it is diagonalisable.

In this case, our vector space is $\mathfrak{g}$ and the map is the automorphism σ' on $\mathfrak{g}$ induced by the automorphism σ on G . Since σ' is an involution, per our remark above we can split $\mathfrak{g}$ into the 1 and -1 eigenspaces of σ' and we note that the 1 eigenspace is exactly $\mathfrak{h}$. We let the -1 eigenspace be $\mathfrak{m}$. It remains to show the three results involving Lie brackets:

1. $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$ follows immediately from the fact that $\mathfrak{h}$ is a subalgebra

2. For $X \in \mathfrak{h}, Y \in \mathfrak{m}$, we have $\sigma'[(X, Y)] = [\sigma'(X), \sigma'(Y)] = [X, -Y] = -[X, Y]$ where the first equality follows from σ' being a Lie algebra automorphism and thus preserving the structure under Lie brackets, the second equality follows from the properties of the eigenspace and the final equality following from the skew-symmetric property of the Lie bracket. Thus $[X, Y]$ is in the -1 eigenspace and is hence in $\mathfrak{m}$.
3. For $A, B \in \mathfrak{m}$, $\sigma'([A, B]) = [-A, -B] = [A, B]$. Thus this is in the 1 eigenspace and hence in $\mathfrak{h}$.

This concludes the proof.

Our final result for this section relates $\mathfrak{m}$ with the tangent space of the coset space G/H :

Theorem 4.25. *Let S be a symmetric space with $S \cong G/H$, and let $\mathfrak{g} = \text{Lie}(G)$, $\mathfrak{h} = \text{Lie}(H)$. Denoting $o = eH$, we have that:*

$$T_o(G/H) \cong \mathfrak{g}/\mathfrak{h} \cong \mathfrak{m}$$

Proof. Consider the projection $\pi : G \rightarrow G/H$ given by $\pi(g) = gH$. Consider its differential at the identity coset $d\pi_e : \mathfrak{g} \rightarrow T_o(G/H)$ (recall that $T_e G \cong \mathfrak{g}$). This projection has kernel H , so naturally the kernel of the differential is $\mathfrak{h}$. By the first isomorphism theorem, we hence have $T_o(G/H) \cong \mathfrak{g}/\mathfrak{h}$. Since $\mathfrak{g}$ is a direct sum of $\mathfrak{h}$ and $\mathfrak{m}$, $\mathfrak{m} \cong \mathfrak{g}/\mathfrak{h}$ and we have the desired result. $\square$

4.5 Non-example of a symmetric space

In this subsection we give an example of a non-symmetric homogeneous space: the automorphism group of the octonions. We first introduce the octonions below.

Definition 4.26 (Octonions). The *octonions*, denoted by $\mathbb{O}$, are a normed division $\mathbb{R}$ -algebra with basis $\{1, e_1, e_2, \dots, e_7\}$.

Remark 4.27 (Octonion multiplication). To define octonion multiplication, we need to define products of the basis vectors. For $e_i e_j$ for $i \neq j$, we can use a Fano plane, depicted in Figure 6, which is nicer to work with than multiplication tables. Note that each line in the plane, including the inscribed circle, goes through exactly three points.

Consider any line in Figure 6 and the points on the line (e_i, e_j, e_k) , ordered in the direction of the line's arrow. Then we have that $e_i e_j = e_k$ and $e_j e_i = -e_k$, so e_i and e_j anti-commute. Moreover these relations hold for cyclic permutations of e_i, e_j, e_k .

We separately define $1e_i = e_i 1 = e_i$ and $e_i^2 = -1$ for all $i = 1, \dots, 7$.

Remark 4.28 (Fun properties of the octonions). From these multiplication rules we immediately see that the octonions are non-commutative, like the quaternions, but also non-associative: $(e_1 e_2) e_4 = e_7$ but $e_1 (e_2 e_4) = -e_7$. By Hurwitz's theorem the octonions are the highest-dimensional normed division algebra over $\mathbb{R}$.

Definition 4.29 (Conjugate). Let $x = x_0 + x_1 e_1 + \dots + x_7 e_7$ be an octonion, then its *conjugate* is defined:

$$\bar{x} = x_0 - x_1 e_1 - \dots - x_7 e_7$$

Note that a quick computation shows that the conjugate of a product $\overline{uv} = \bar{v}\bar{u}$.

Similarly to complex numbers and quaternions, we can split octonions into a real part spanned by 1 , and an imaginary part spanned by $e_1, \dots, e_7$:

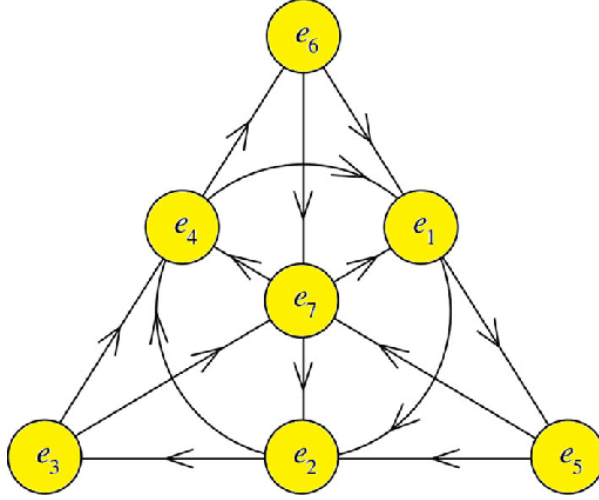


Figure 6: Fano plane for the octonions (Gillard and Gresnigt, 2019).

Definition 4.30 (Real and imaginary parts). Let x be an octonion. Then we define the *real part* $\operatorname{Re} x := \frac{x+\bar{x}}{2}$ and the *imaginary part* $\operatorname{Im} x := \frac{x-\bar{x}}{2}$. x is an *imaginary octonion* if $\operatorname{Re} x = 0$.

By an octonion automorphism A we mean that $A : \mathbb{O} \rightarrow \mathbb{O}$ is an invertible $\mathbb{R}$ -linear map such that $A(xy) = (Ax)(Ay)$ for all $x, y \in \mathbb{O}$. These automorphisms form a group under composition, denoted G_2 , which induces a group action on $\mathbb{O}$.

The octonion inner product is defined in the same way as the standard Hermitian form, specifically $(u, v) := \operatorname{Re} u\bar{v}$, and the octonion norm is defined $|x|^2 := (x, x) = x\bar{x}$ as usual. As imaginary octonions correspond with vectors in $\mathbb{R}^7$, we can identify the set of unit norm imaginary octonions with $\mathbb{S}^6$; to show that G_2 acts on $\mathbb{S}^6$ as well, we prove below that automorphisms preserve the inner product:

Lemma 4.31. *Let A denote an octonion automorphism, then $(Ax, Ay) = (x, y)$ for all $x, y \in \mathbb{O}$.*

Proof. $A1 = A(1 \cdot 1) = (A1)(A1)$ and $A1 \neq 0$ as A is injective, hence $A1 = 1$. Moreover $(Ae_i)^2 = A(e_i^2) = -1$, so

$$-Ae_i = (Ae_i)^{-1} = \frac{\overline{Ae_i}}{|Ae_i|^2} = \frac{\overline{Ae_i}}{|-1|} = \overline{Ae_i}$$

Using these equalities it can be shown that $\overline{Ax} = A\bar{x}$ for all $x \in \mathbb{O}$, and the result can be proven from here. $\square$

Theorem 4.32. *G_2 is a Lie group.*

Proof. G_2 is a subgroup of $O(8)$ by the previous lemma. Consider an arbitrary sequence $A_n \subset G_2$ which converges to $A \in O(8)$. We check A respects multiplication:

$$A(xy) = \lim_{n \rightarrow \infty} A_n(xy) = \left(\lim_{n \rightarrow \infty} A_n x \right) \left(\lim_{n \rightarrow \infty} A_n y \right) = (Ax)(Ay)$$

for all $x, y \in \mathbb{O}$, showing that $A \in G_2$, hence G_2 is closed. G_2 is a closed subgroup of the Lie group $O(8)$, hence a Lie subgroup by the Closed Subgroup Theorem from Lee, 2002. $\square$

Using Lemma 4.31 and the previous theorem, it isn't hard to see that $G_2 \times \mathbb{S}^6 \rightarrow \mathbb{S}^6$ mapping $A \cdot x = Ax$ is indeed an action. Moreover this is a transitive action as G_2 acts transitively on $\mathbb{O}$, and by identifying octonion automorphisms as matrices in $O(8)$, we can see the action is smooth too. Now we compute the isotropy subgroup of e_1 .

Definition 4.33 (Simply transitive). Let G be a group action on a set X . G is a *simply transitive* action if for all $x, y \in X$, there exists a unique $g \in G$ such that $g \cdot x = y$.

Lemma 4.34. *Let u, v be imaginary octonions. Then u, v anti-commute if and only if they are orthogonal.*

Proof. Since they are imaginary, $\bar{u}, \bar{v} = -u, -v$, thus:

$$(u, v) = \operatorname{Re} u\bar{v} = \frac{1}{2}(u\bar{v} + v\bar{u}) = \frac{1}{2}(-uv - vu)$$

u, v are orthogonal when their inner product is zero, which is equivalent to them anti-commuting according to the above expression. $\square$

An important property of the octonions is that they can be generated using specific triples:

Remark 4.35. A *Cayley triple* (u_1, u_2, u_3) is a triple of unit imaginary octonions such that u_1 anti-commutes with u_2 and u_3 anti-commutes with u_1, u_2 and $u_1 u_2$. In particular a Cayley triple generates the octonions.

We use the previous results to show $(G_2)_{e_1} = \operatorname{SU}(3)$, with a proof based on one in Daura Serrano et al., 2022:

Proposition 4.36. *The isotropy subgroup $(G_2)_{e_1}$ is $\operatorname{SU}(3)$.*

Proof. Let $z = z_0 + z_1 e_1 + \dots + z_7 e_7$ denote an arbitrary octonion. Identifying the space spanned by $1, e_1$ with the complex numbers, we can use the multiplication rules of the octonions to rearrange z into:

$$z = w_0 + w_1 e_2 + w_2 e_3 + w_3 e_5 \quad (7)$$

where $w_0, w_1, w_2, w_3 \in \mathbb{C}$. We can see that $z \in \mathbb{C} \oplus \mathbb{C}^3$, which gives an isomorphism $\alpha : \mathbb{O} \rightarrow \mathbb{C} \oplus \mathbb{C}^3$ such that $z \mapsto w_0 + (w_1, w_2, w_3)$. T fixes all complex numbers by the isotropy property, so T is wholly and uniquely characterised by where it maps e_2, e_3, e_5 to:

$$Tz = w_0 + w_1 T e_2 + w_2 T e_3 + w_3 T e_5 \quad (8)$$

$\mathbb{C}^3 = \langle e_2, e_3, e_5 \rangle$ is invariant under T : $(\alpha \circ T)(e_2) = w_0 + (w_1, w_2, w_3)$ with $w_0 = 0$ as e_2 is an imaginary octonion, so $T e_2 \in \mathbb{C}^3$, and the same is true for e_3 and e_5 . Hence we can construct a matrix $A = [T]_B \in M(3, \mathbb{C})$ where B denotes the basis $\{e_2, e_3, e_5\}$, and since T is invertible it follows that $A \in \operatorname{GL}(3, \mathbb{C})$. By Lemma 4.31 T preserves the octonion inner product, and the latter induces the standard inner product on $\mathbb{R}^6 \cong \mathbb{C}^3$ so the corresponding matrix $A \in \operatorname{SO}(6)$, hence $(G_2)_{e_1} \subseteq \operatorname{U}(3)$.

Now we shall show a one-to-one correspondence between orthonormal $\mathbb{C}^3$ vectors and Cayley triples. For this we will need to rewrite the octonion inner product: let $u = u_1 e_2 + u_2 e_3 + u_3 e_5$ and $v = v_1 e_2 + v_2 e_3 + v_3 e_5$ be octonions in $\mathbb{C}^3$, then:

$$\begin{aligned} (u, v) &= \frac{1}{2}(u\bar{v} + v\bar{u}) = \frac{1}{2}(u_1 \bar{v}_1 + u_2 \bar{v}_2 + u_3 \bar{v}_3 + v_1 \bar{u}_1 + v_2 \bar{u}_2 + v_3 \bar{u}_3) \\ &= \frac{1}{2}(\langle u, v \rangle + \langle v, u \rangle) = \frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle}) = \operatorname{Re} \langle u, v \rangle \end{aligned}$$

where $\langle \cdot, \cdot \rangle$ denotes the standard Hermitian inner product in $\mathbb{C}^3$. Let $u, v \in \mathbb{C}^3$ be orthogonal in $\mathbb{C}^3$: we see immediately from above that they are orthogonal in the octonions so they anti-commute via Lemma 4.34. Similarly u and v are both orthogonal to e_1 , so both anti-commute

with e_1 . Finally $e_1 u = iu \in \mathbb{C}^3$ and $\langle iu, v \rangle = i\langle u, v \rangle = 0$ so $e_1 u$ is orthogonal and thus anti-commutes with v . All this shows that (e_1, u, v) forms a Cayley triple.

In the opposite direction, assume (e_1, u, v) is a Cayley triple. Then $\langle u, v \rangle = \operatorname{Re}\langle u, v \rangle = 0$ and

$$(e_1 u, v) = \operatorname{Re}\langle iu, v \rangle = \operatorname{Re} i\langle u, v \rangle = \operatorname{Im}\langle u, v \rangle = 0$$

so $\langle u, v \rangle = 0$ and u, v are orthonormal in $\mathbb{C}^3$. The consequence of this: Cayley triples generate the octonions so $(G_2)_{e_1}$ acts simply transitively on Cayley triples, but we have shown that the triples correspond with pairs of orthonormal $\mathbb{C}^3$ vectors, giving us a simply transitive action of $(G_2)_{e_1}$ on these orthonormal pairs.

Consider an arbitrary basis of orthonormal eigenvectors of A in $\mathbb{C}^3$, which exists by the spectral theorem. Then consider the first two basis vectors v_1, v_2 ; there exists a unique $\phi \in (G_2)_{e_1}$ such that $\phi \cdot (v_1, v_2) = (e_2, e_3)$ by simple transitivity. Then e_2, e_3 are orthonormal eigenvectors of $B = \phi A \phi^{-1}$, and denoting their eigenvalues λ and μ , it can be shown that e_5 is the third orthonormal basis vector of B with eigenvalue $\overline{\lambda\mu}$. $\det A = \det B = \lambda\mu\overline{\lambda\mu} \geq 0$ so $A \in \operatorname{SU}(3)$. Hence we finally have $(G_2)_{e_1} \subseteq \operatorname{SU}(3)$.

The opposite direction is simpler. Let $A \in \operatorname{SU}(3)$, then by Equation 8 there is a unique octonion automorphism $T \in (G_2)_{e_1}$ such that $Te_k = Ae_k$ for $k = 2, 3, 5$. $\square$

This shows that $\mathbb{S}^6 \cong G_2/\operatorname{SU}(3)$ is a homogeneous space.

Assuming that this is a symmetric space, it follows from Theorem 4.23 that the Lie algebra $\mathfrak{g}_2$ can be decomposed into the direct sum $\mathfrak{su}_3 \oplus \mathfrak{m}$ such that $\mathfrak{m}$ is isomorphic to $T_0 G_2/\operatorname{SU}(3) \cong \mathfrak{g}_2/\mathfrak{su}_3$, where we denote the identity coset $\operatorname{Id}_{\mathbb{O}}\operatorname{SU}(3)$ with 0.

Since G_2 is 14-dimensional (Yokota, 2025) and $\operatorname{SU}(3)$ is 8-dimensional, $\mathfrak{m}$ is a 6-dimensional vector space, hence it is isomorphic to $\mathbb{C}^3 \subset \mathbb{R}^7$. Then $\mathbb{R}^7$ induces a cross product Lie bracket on $\mathfrak{m}$; in particular, by closure of the cross product this implies that $[\mathfrak{m}, \mathfrak{m}] \not\subset \mathfrak{su}_3$, contradicting Theorem 4.23.

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A Axioms of a manifold

Here are some examples of topological spaces which are locally Euclidean, but not Hausdorff or second countable.

A.1 Counterexample for Hausdorff

Line with two origins: Consider two copies of $\mathbb{R}$: $\mathbb{R}_1 = \mathbb{R} \times \{0\}$, $\mathbb{R}_2 = \mathbb{R} \times \{1\}$ and the quotient space $X = (\mathbb{R}_1 \cup \mathbb{R}_2)/\sim$, where $\sim$ identifies $x \times \{0\}$ with $x \times \{1\}$ if $x \neq 0$. The topology we put on the space is by taking the finest topology for which the projection map

$$\pi : \mathbb{R}_1 \cup \mathbb{R}_2 \rightarrow X \text{ via } x \mapsto [x]$$

is continuous. More explicitly:

$$U \text{ is open in } X \iff \pi^{-1}(U) \text{ is open in } \mathbb{R}_1 \cup \mathbb{R}_2$$

This is locally Euclidean of dimension 1: for every element $x \in X$ you can take an open set U containing x such that $\pi^{-1}(U)$ is two copies of open intervals. X is second countable as $\mathbb{R}_1 \cup \mathbb{R}_2$ is second countable, and π is an open mapping.

X is not Hausdorff as you can't separate the two origins: for any $U \subset X$ containing the origin $\{0\} \times \{0\}$, the preimage $\pi^{-1}(U)$ contains some open interval $(-\epsilon, \epsilon)$ in $\mathbb{R}_1$. Similarly, for any open set $V \subset X$ containing the other origin $\{0\} \times \{1\}$, $\pi^{-1}(V)$ will contain some interval $(-\eta, \eta)$. Then $U \cap V \neq \emptyset$.

A.2 Counterexamples for second countability

1. *Long Line:* Constructed by extending the real line using the first uncountable ordinal ω_1 , the long line is Hausdorff and locally Euclidean but lacks a countable basis due to its uncountable “length”.
2. *Spaces with too many pieces:* Consider an uncountable disjoint union of copies of the real line, each with the standard topology. Such a space has no countable basis because the basis must include open sets from uncountably many disconnected parts.

Thus, both examples fail the second countability condition.

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